Backplane

BACKPLANE

File: backplane/backplane.kicad_sch

FLUXO DE DADOS

PRANCHA I: PLACA DOS REGISTRADORES

Placas de registradores:
* FR1-1 (bits pares)
* FR2-2 (bits ímpares)

File: FR1-1/FR1-1.kicad_sch

PRANCHA II: CIRCUITO - UNIDADE ARITMÉTICA

Placas de somador:
* FS1-3 (bits menos significativos)
* FS2-4 (bits mais significativos)

File: FS1-3/FS1-3.kicad_sch

PRANCHA III: ACUMULADOR

Placa de acumulador:
* FAC-5

File: FAC-5/FAC-5.kicad_sch

PRANCHA IV: FLUXO DE DADOS

W.I.P.

Placa:
* FM1-6 (mais-um e porta de seleção 6)

File: FM1-6/FM1-6.kicad_sch

PRANCHA V: PLACA CONTROLE DA MEMÓRIA

Placa:
* FCM-7 (interface com memória)

File: FCM-7/FCM-7.kicad_sch

PRANCHA VI: DADOS PARA CARTÕES INTERFACE

Placa:
* FIN-8 (sinais para cartões de interface)

File: FIN-8/FIN-8.kicad_sch

PAINEL

PRANCHA VII: PLACA MOSTRA REGISTRADORES

Placa:
* PMR-1 (drivers para os LEDs)

File: PMR-1/PMR-1.kicad_sch

PRANCHA VIII: PLACA CHAVES E PILOTO

Placa:
* PCP-2 ("debounce" para chaves e drivers para lâmpadas piloto)

File: PCP-2/PCP-2.kicad_sch

UNIDADE DE CONTROLE

PRANCHA IX: RELÓGIO CENTRAL

Placa:
* CRC-1

File: CRC-1/CRC-1.kicad_sch

PRANCHA X: DECODIFICADOR

Placa:
* CDE-3

File: CDE-3/CDE-3.kicad_sch

PRANCHA XI: CONTROLE DE ESTADO

W.I.P.

Placa:
* CCE-2

File: CCE-2/CCE-2.kicad_sch

PRANCHA XII: PLACA CONTROLE 1

Placa:
* CT1-4

File: CT1-4/CT1-4.kicad_sch

PRANCHA XIII: PLACA SINAIS CONTROLE

W.I.P.

Placa:
* CT2-5

File: CT2-5/CT2-5.kicad_sch

PRANCHA XIV: PLACA DE CONTROLE

W.I.P.

Placa:
* CT3-6

File: CT3-6/CT3-6.kicad_sch

PRANCHA XV: SINAIS DE CONTROLE

Placa:
* CT4-7

File: CT4-7/CT4-7.kicad_sch

PRANCHA XVI: SINAIS DE CONTROLE

Placa:
* CT5-8

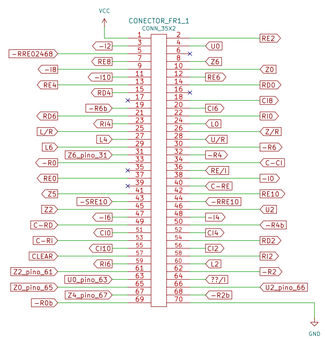
File: CT5-8/CT5-8.kicad_sch

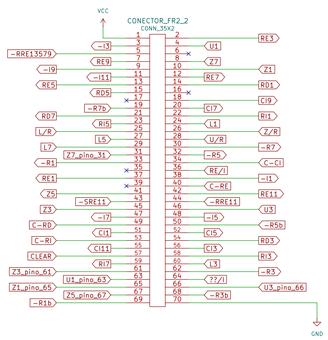
Re-desenho: Felipe Sanches
Projeto: Edson Fregni
Sinais de Controle
Computador de Controle - Patinho Feio
EPUSP LSD

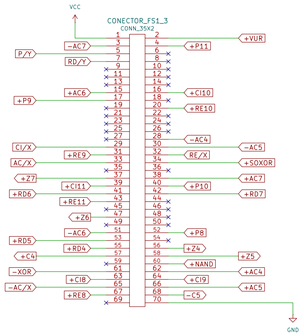
Sheet: /
File: patinhofeio.kicad_sch

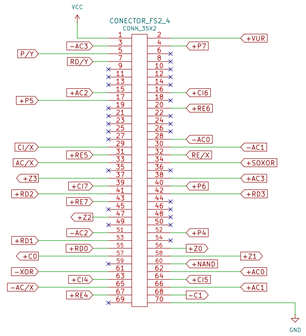
Title: PRANCHA XVI

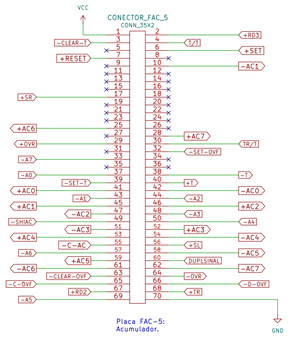
Size: A4	Date: 2 aug 2016	Rev:
KiCad E.D.A. 10.0.3		Id: 1/18





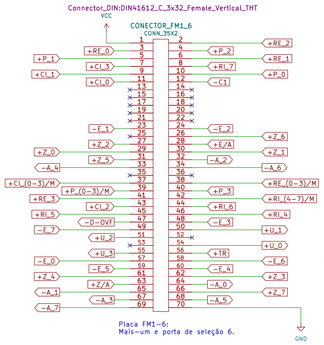




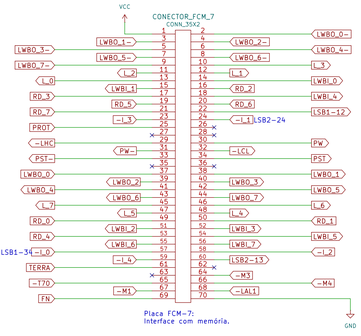


Placa FAC-5:
Acumulador.

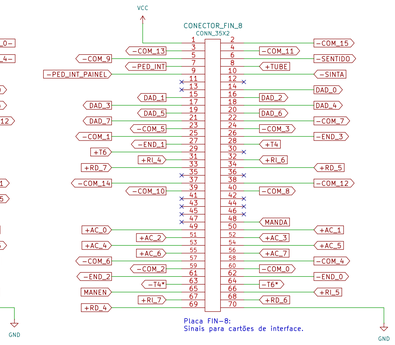
Connector_DIN:DIN41612_C_3x32_Female_Vertical_THT



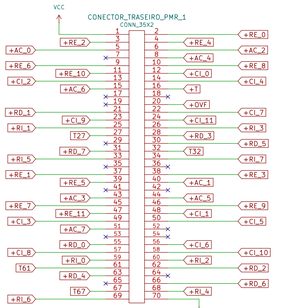
Placa FM1-6:
Mais-um e porta de seleção 6.



Placa FCM-7:
Interface com memória.



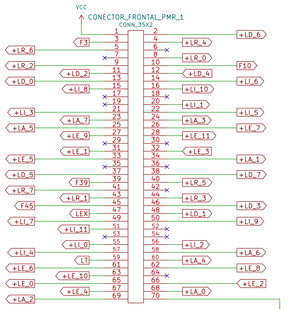
Placa FIN-8:
Sinais para cartões de interface.



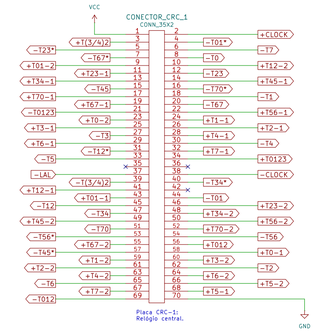
CONECTOR TRASEIRO

PMR-1: PLACA MOSTRA REGISTRADORES

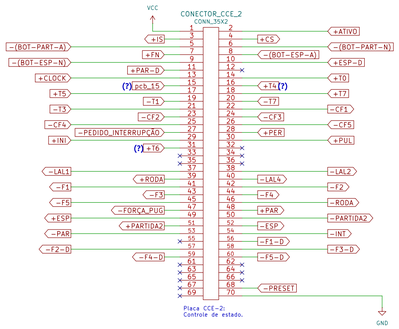
Drivers para os LEDs.



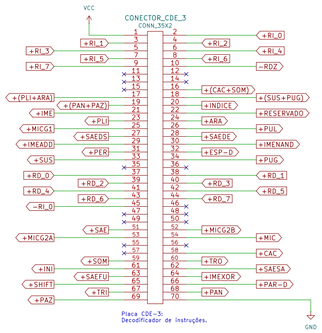
CONECTOR FRONTAL



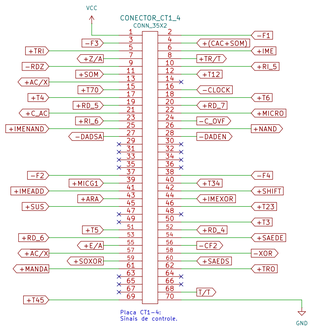
Placa CRC-1:
Relógio central.



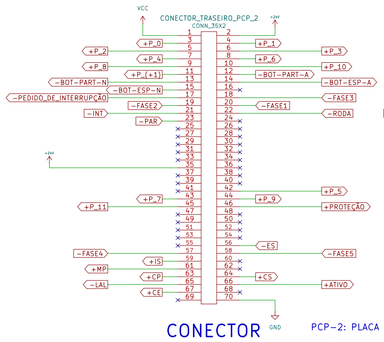
Placa CCE-2:
Controle de estado.



Placa CDE-3:
Decodificador de instruções.



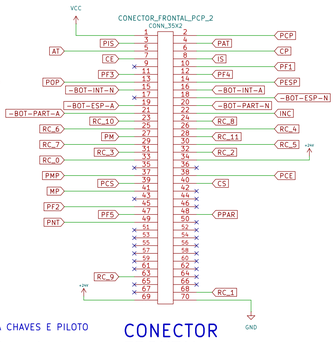
Placa CT1-4:
Sinais de controle.



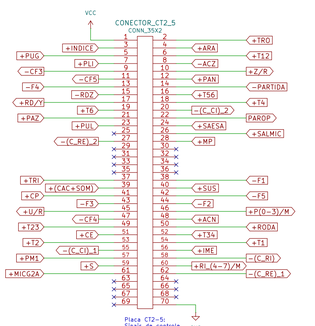
CONECTOR TRASEIRO

PCP-2: PLACA CHAVES E PILOTO

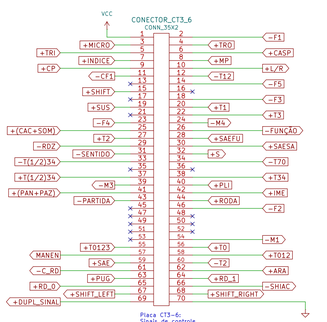
"Debounce" para chaves e drivers para lâmpadas piloto.



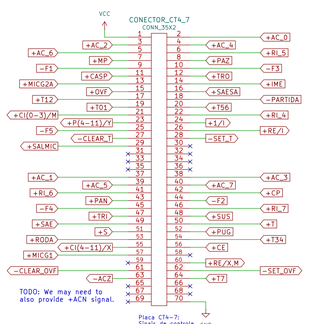
CONECTOR FRONTAL



Placa CT2-5:
Sinais de controle.

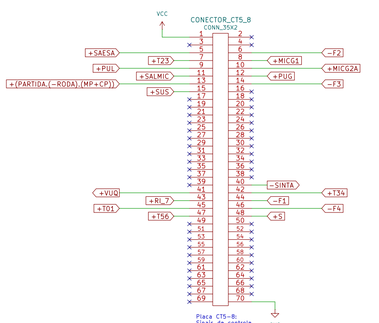


Placa CT3-6:
Sinais de controle.



TODO: We may need to also provide +ACN signal.

Placa CT4-7:
Sinais de controle.

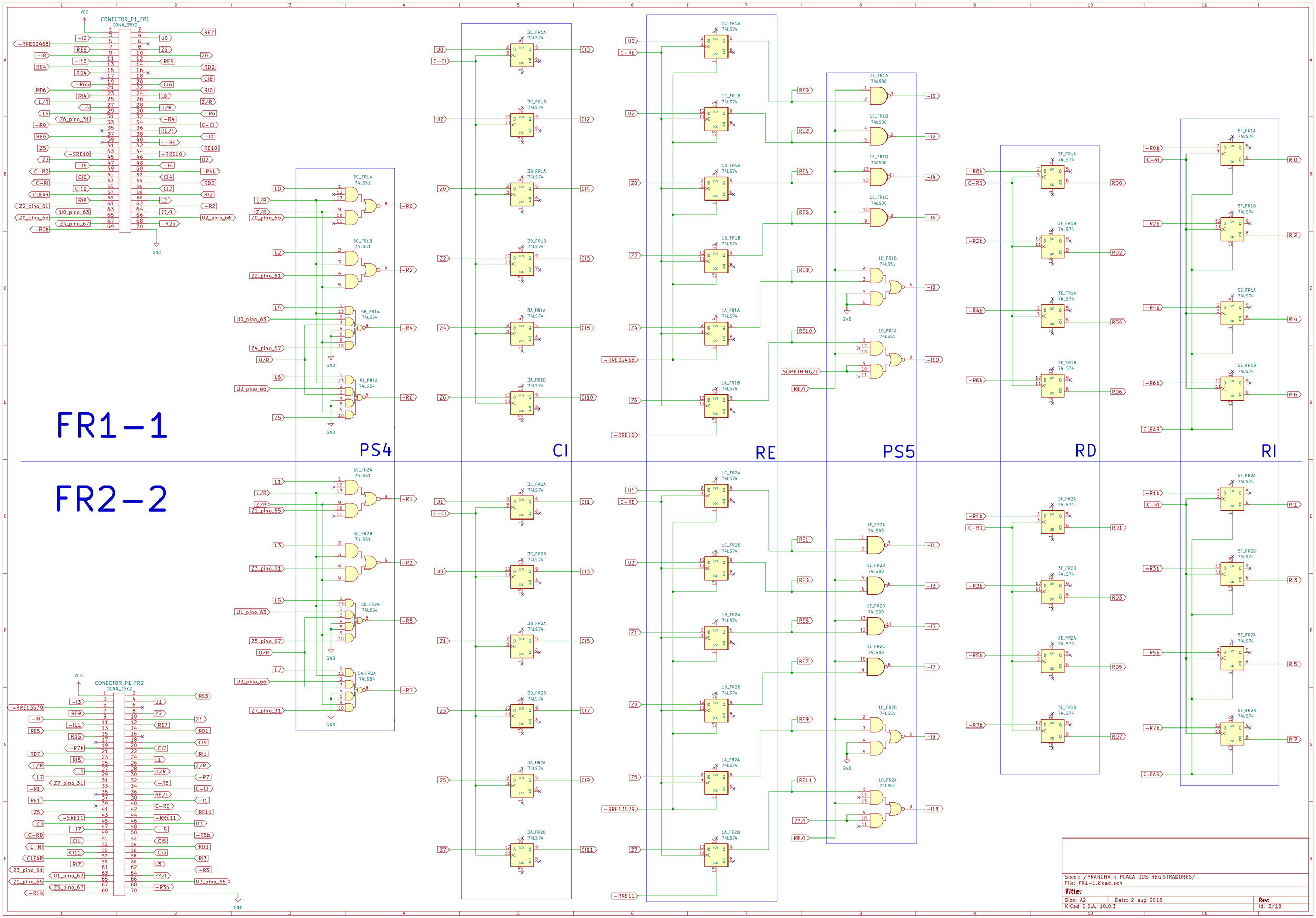


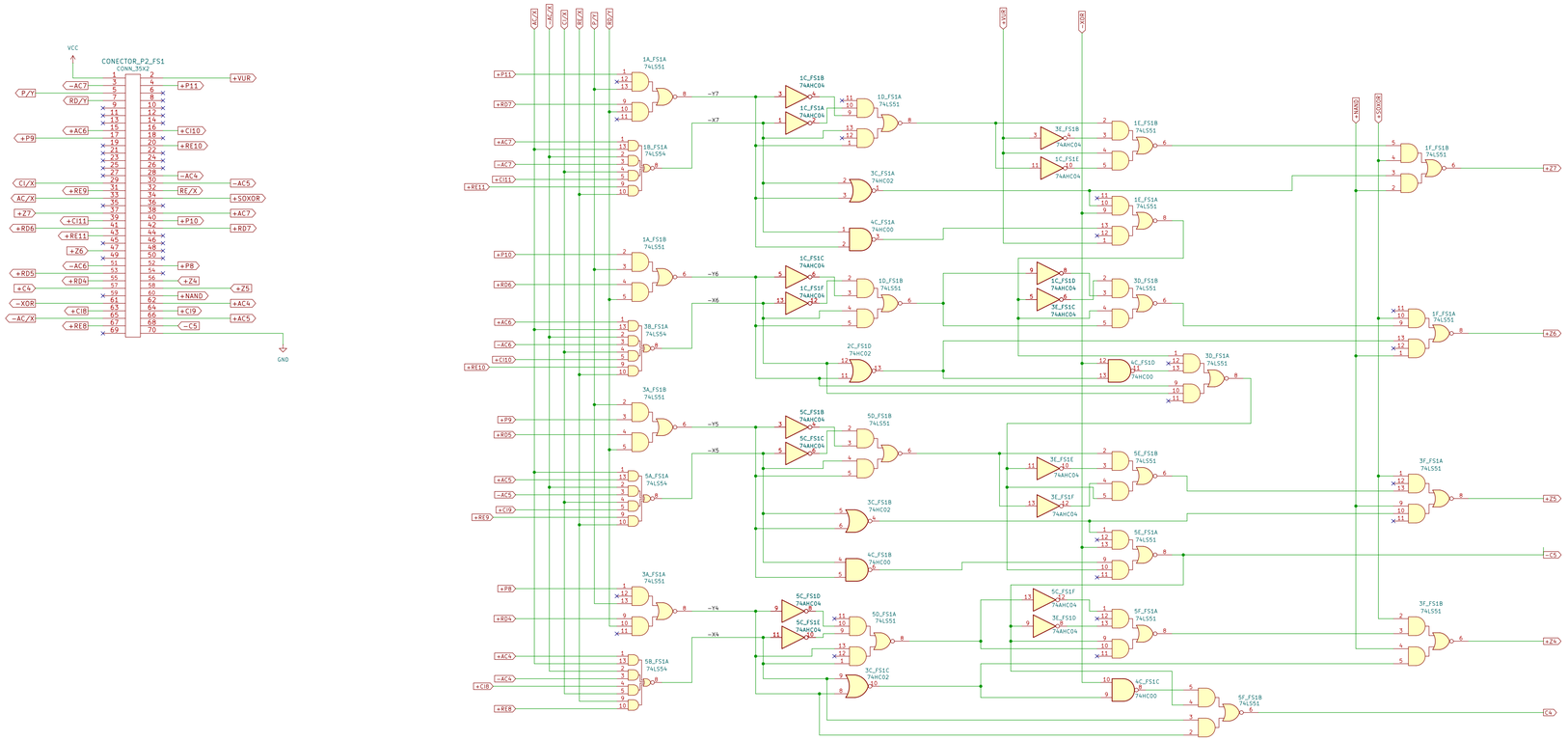
Placa CT5-8:
Sinais de controle.

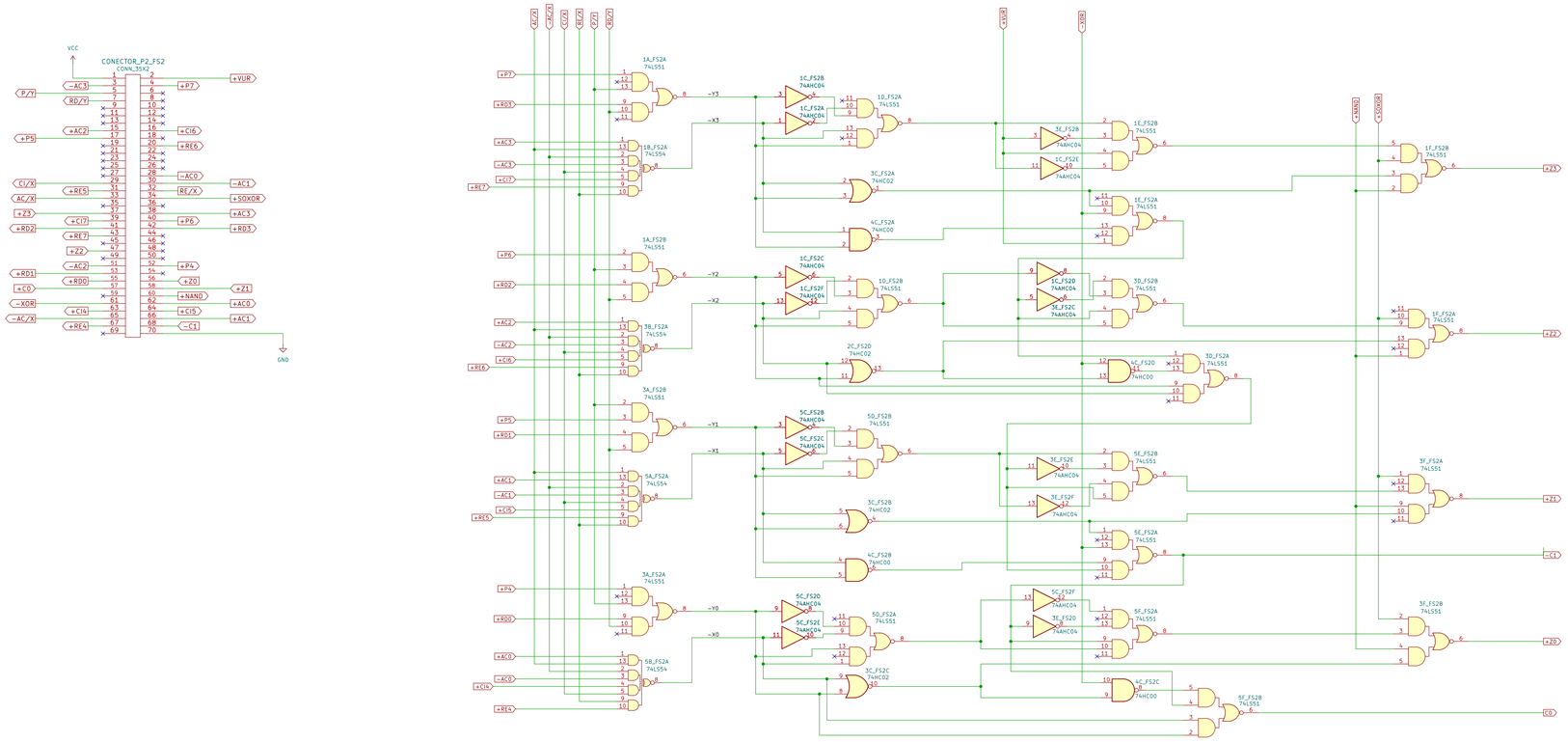
Sheet: /Backplane/
File: backplane.kicad_sch

Title:

Size: A1	Date:	Rev:
KiCad E.D.A. 10.0.3		Id: 2/18


FR1-1
FR2-2
CONECTOR_P1_FR1
CONN_35X2
CONECTOR_P1_FR2
CONN_35X2
PS4
CI
RE
PS5
RD
RI
5C_FR1A 74LS51
5C_FR1B 74LS51
5B_FR1A 74LS54
5A_FR1A 74LS54
5C_FR2A 74LS51
5C_FR2B 74LS51
5B_FR2A 74LS54
5A_FR2A 74LS54
3C_FR1A 74LS74
3C_FR1B 74LS74
3B_FR1A 74LS74
3B_FR1B 74LS74
3A_FR1A 74LS74
3A_FR1B 74LS74
3C_FR2A 74LS74
3C_FR2B 74LS74
3B_FR2A 74LS74
3B_FR2B 74LS74
3A_FR2A 74LS74
3A_FR2B 74LS74
1C_FR1A 74LS74
1C_FR1B 74LS74
1B_FR1A 74LS74
1B_FR1B 74LS74
1A_FR1A 74LS74
1A_FR1B 74LS74
1C_FR2A 74LS74
1C_FR2B 74LS74
1B_FR2A 74LS74
1B_FR2B 74LS74
1A_FR2A 74LS74
1A_FR2B 74LS74
1E_FR1A 74LS00
1E_FR1B 74LS00
1E_FR1D 74LS00
1E_FR1C 74LS00
1D_FR1B 74LS51
1D_FR1A 74LS51
1E_FR2A 74LS00
1E_FR2B 74LS00
1E_FR2D 74LS00
1E_FR2C 74LS00
1D_FR2B 74LS51
1D_FR2A 74LS51
3F_FR1A 74LS74
3F_FR1B 74LS74
3E_FR1A 74LS74
3E_FR1B 74LS74
3F_FR2A 74LS74
3F_FR2B 74LS74
3E_FR2A 74LS74
3E_FR2B 74LS74
5F_FR1A 74LS74
5F_FR1B 74LS74
5E_FR1A 74LS74
5E_FR1B 74LS74
5F_FR2A 74LS74
5F_FR2B 74LS74
5E_FR2A 74LS74
5E_FR2B 74LS74
VCC
GND
Sheet: /PRANCHA I: PLACA DOS REGISTRADORES/
File: FR1-1.kicad_sch
Title:
Size: A2
Date: 2 aug 2016
Rev:
KiCad E.D.A. 10.0.3
Id: 3/18

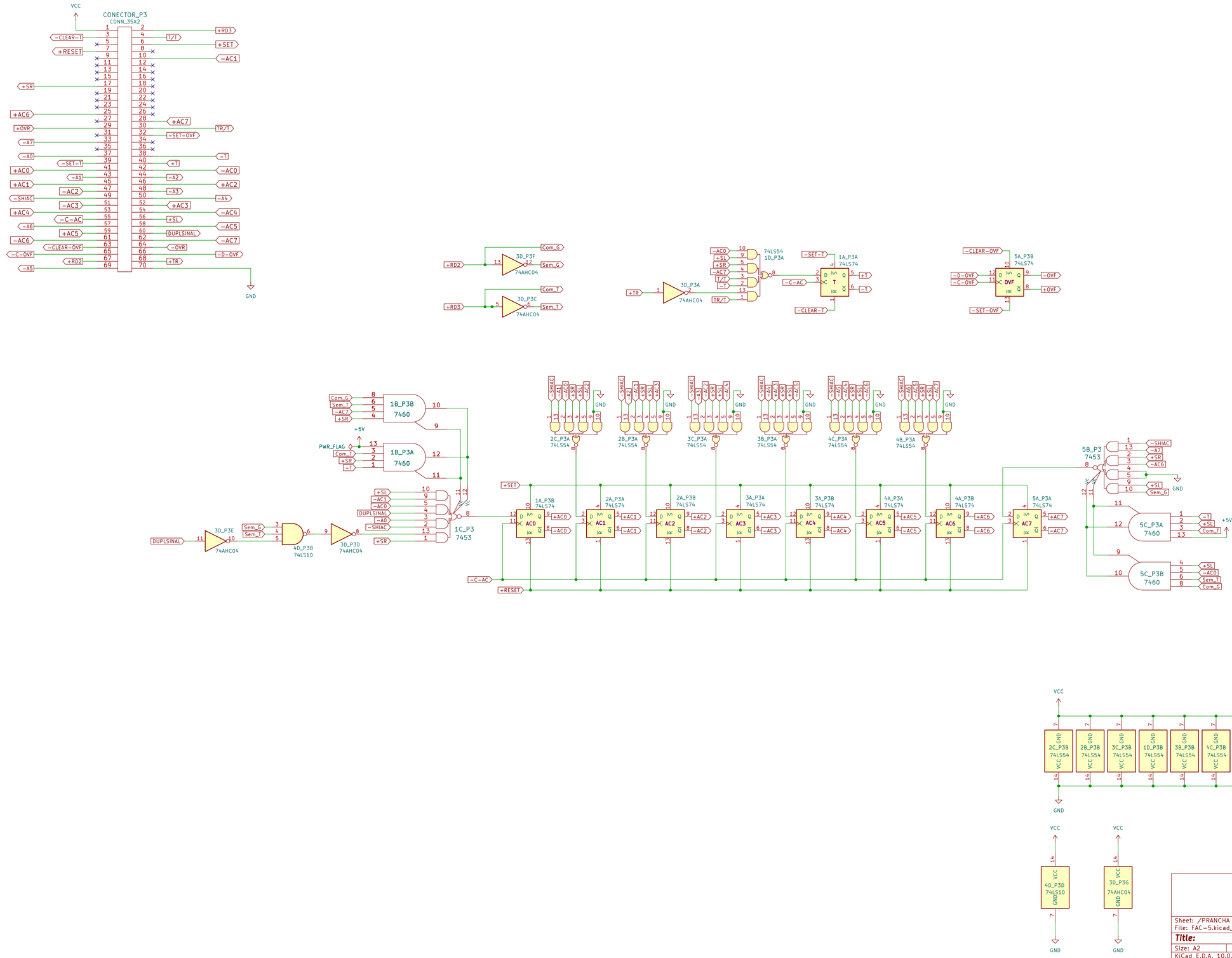




Sheet: /PRANCHA II: CIRCUITO – UNIDADE ARITMÉTICA/
File: FS1-3.kicad_sch

Title:

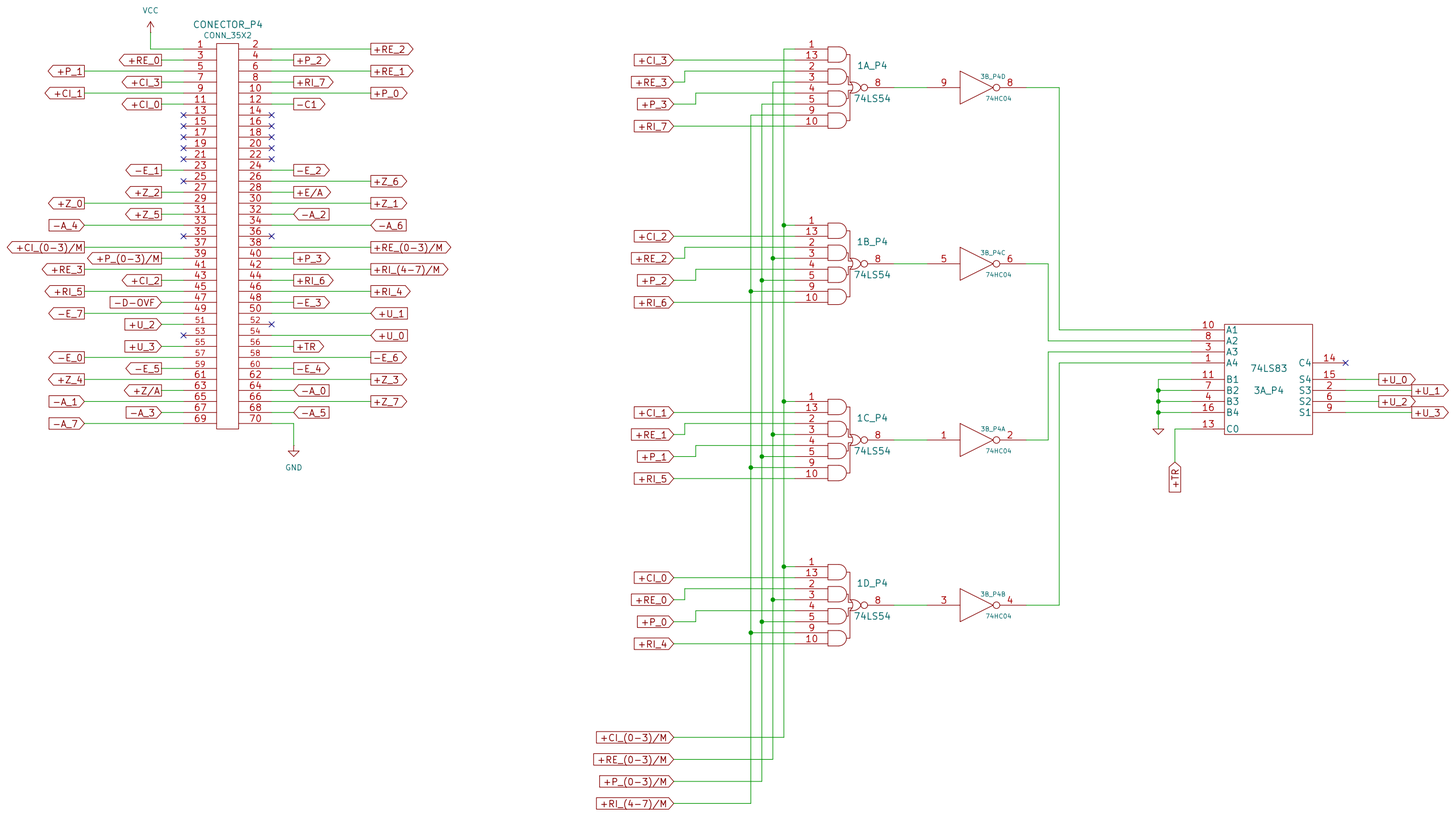
Size: A1	Date: 2026-06-29	Rev:
KiCad E.D.A. 10.0.3		Id: 4/18

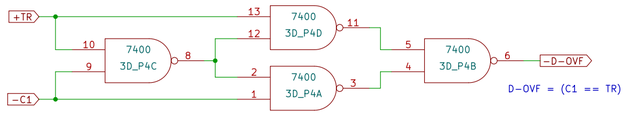


Sheet: /PRANCHA III: ACUMULADOR/
File: FAC-5.kicad_sch

Title:

Size: A2	Date: 2 aug 2016	Rev:
KiCad E.D.A. 10.0.3		Id: 5/18

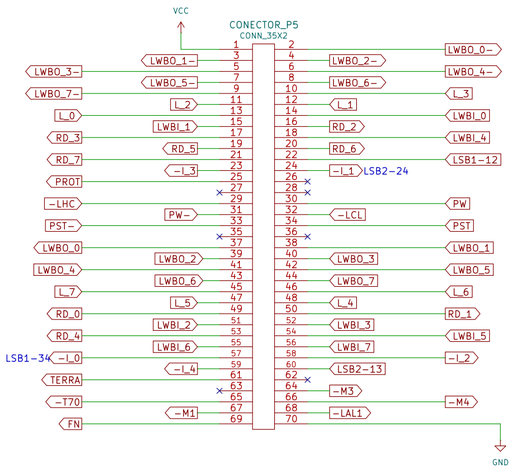


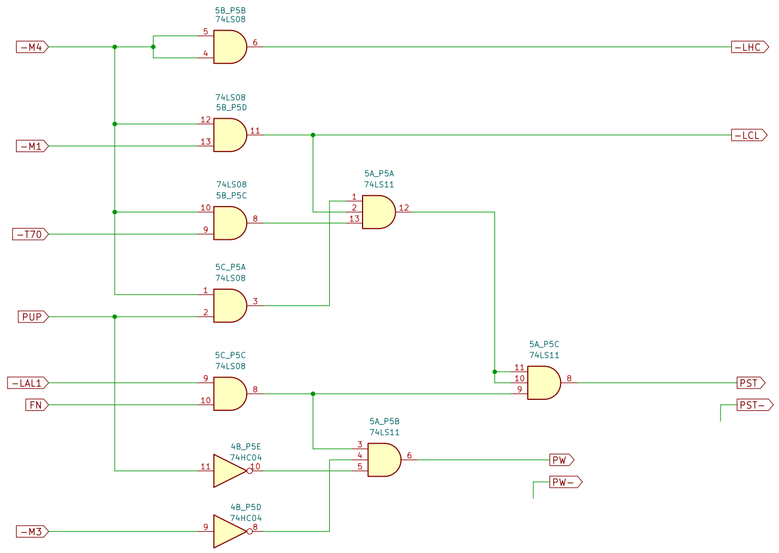


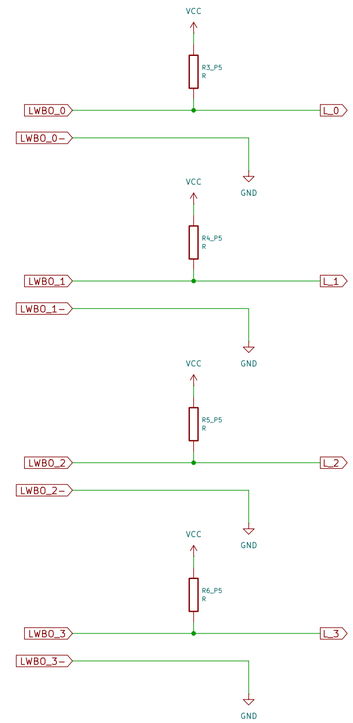
Sheet: /PRANCHA IV: FLUXO DE DADOS/
File: FM1-6.kicad_sch

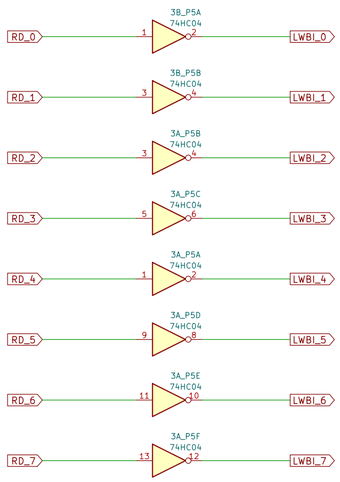
Title: MAIS UM E PORTA DE SELEÇÃO

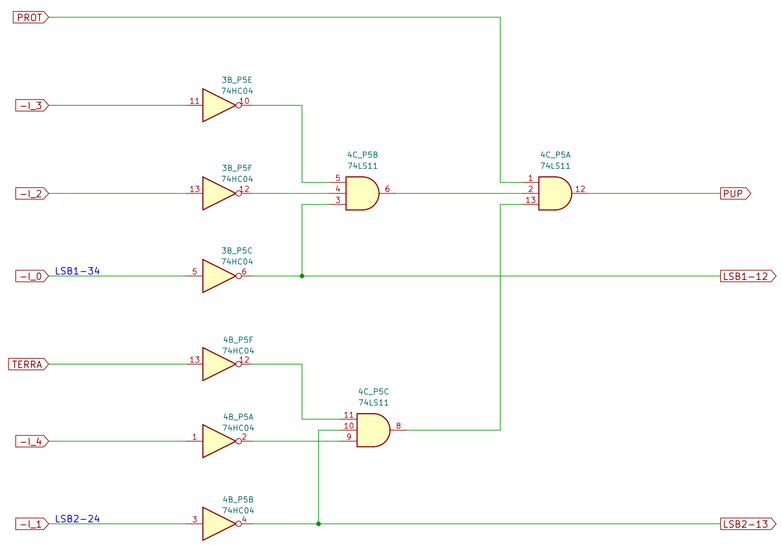
Size: A2	Date: 2 aug 2016	Rev:
KiCad E.D.A. 10.0.3		Id: 6/18

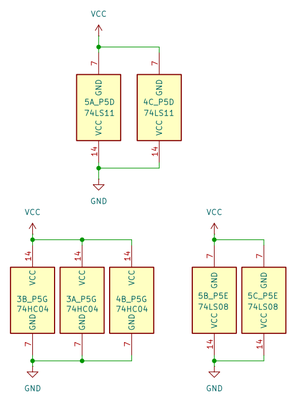








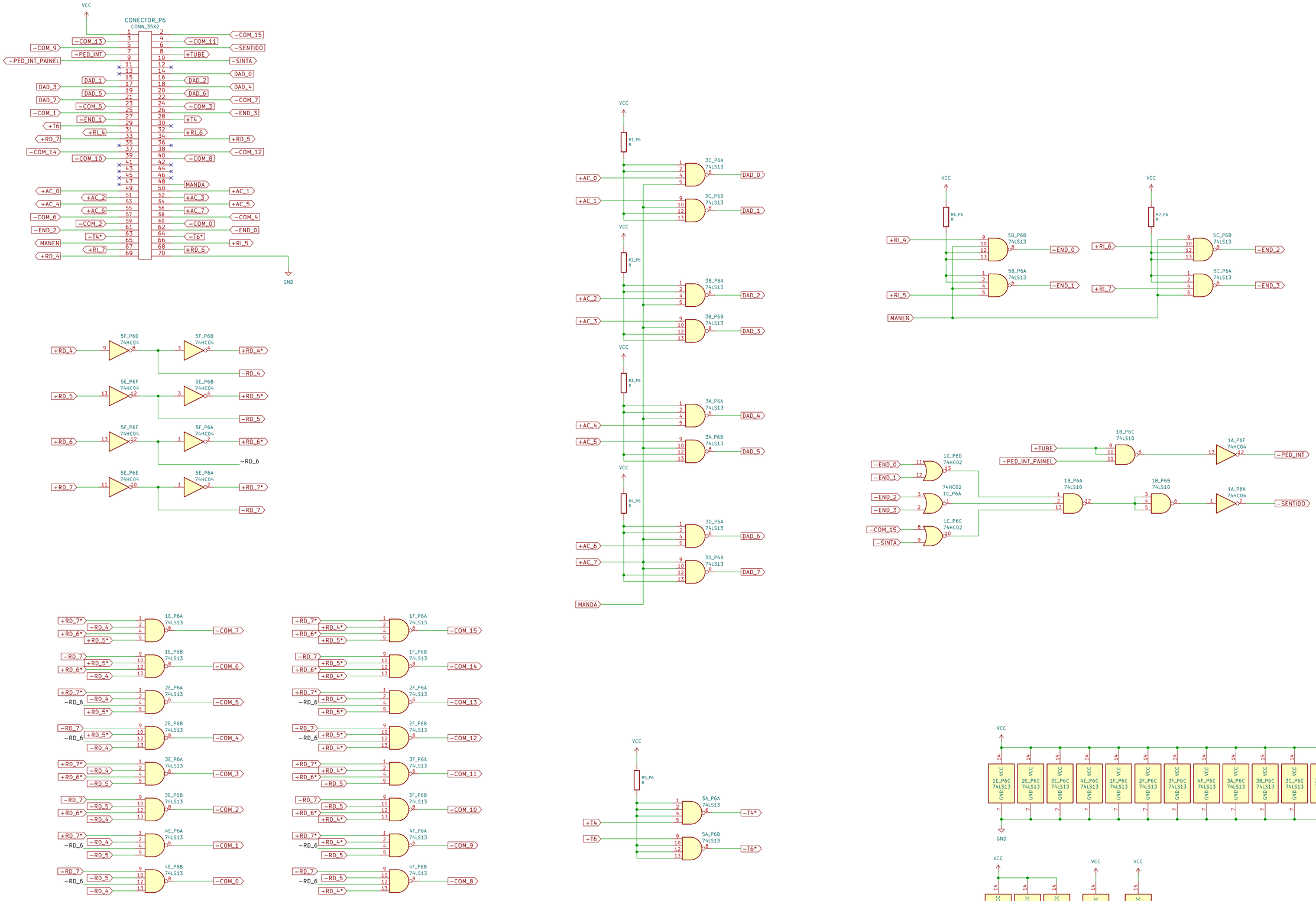




Sheet: /PRANCHA V: PLACA CONTROLE DA MEMÓRIA/
File: FCM-7.kicad_sch

Title:

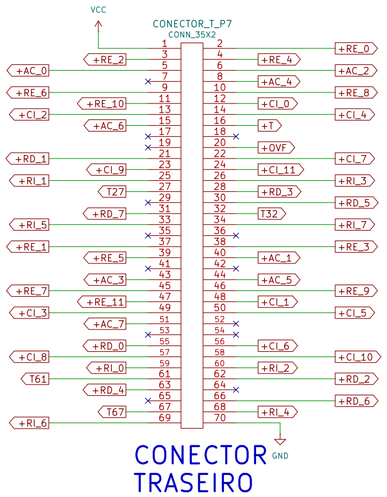
Size: A2	Date: 2 aug 2016	Rev:
KiCad E.D.A. 10.0.3		Id: 7/18

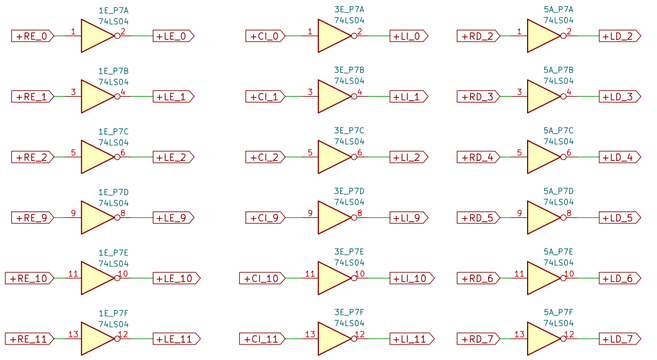


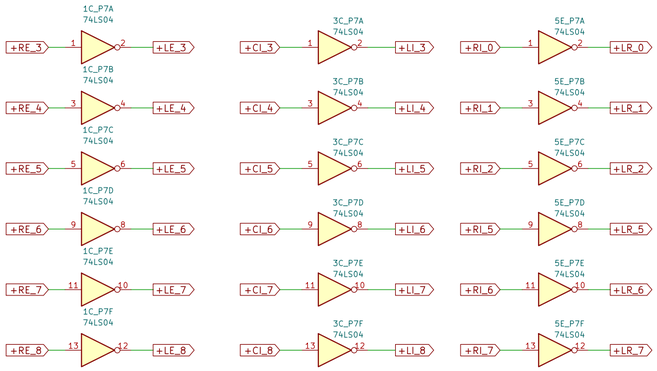
Sheet: /PRANCHA VI: DADOS PARA CARTÕES INTERFACE/
File: FIN-8.kicad_sch

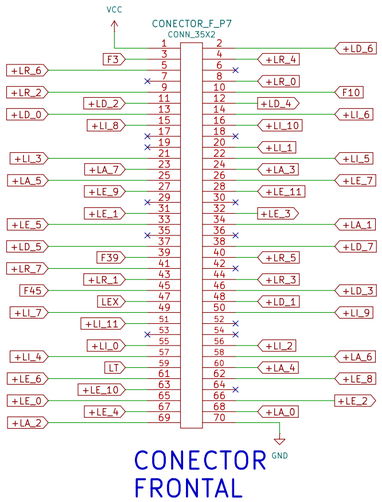
Title:

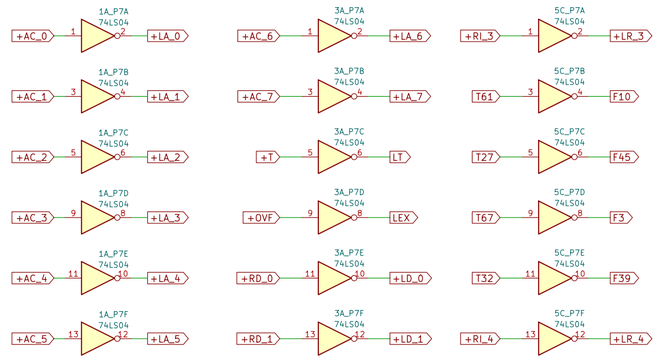
Size: A2	Date: 2 aug 2016	Rev:
KiCad E.D.A. 10.0.3		Id: 8/18

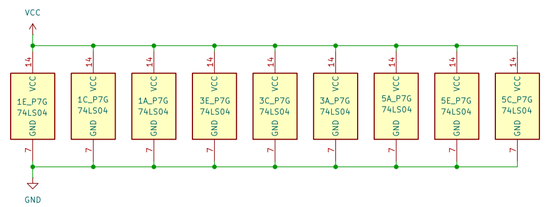








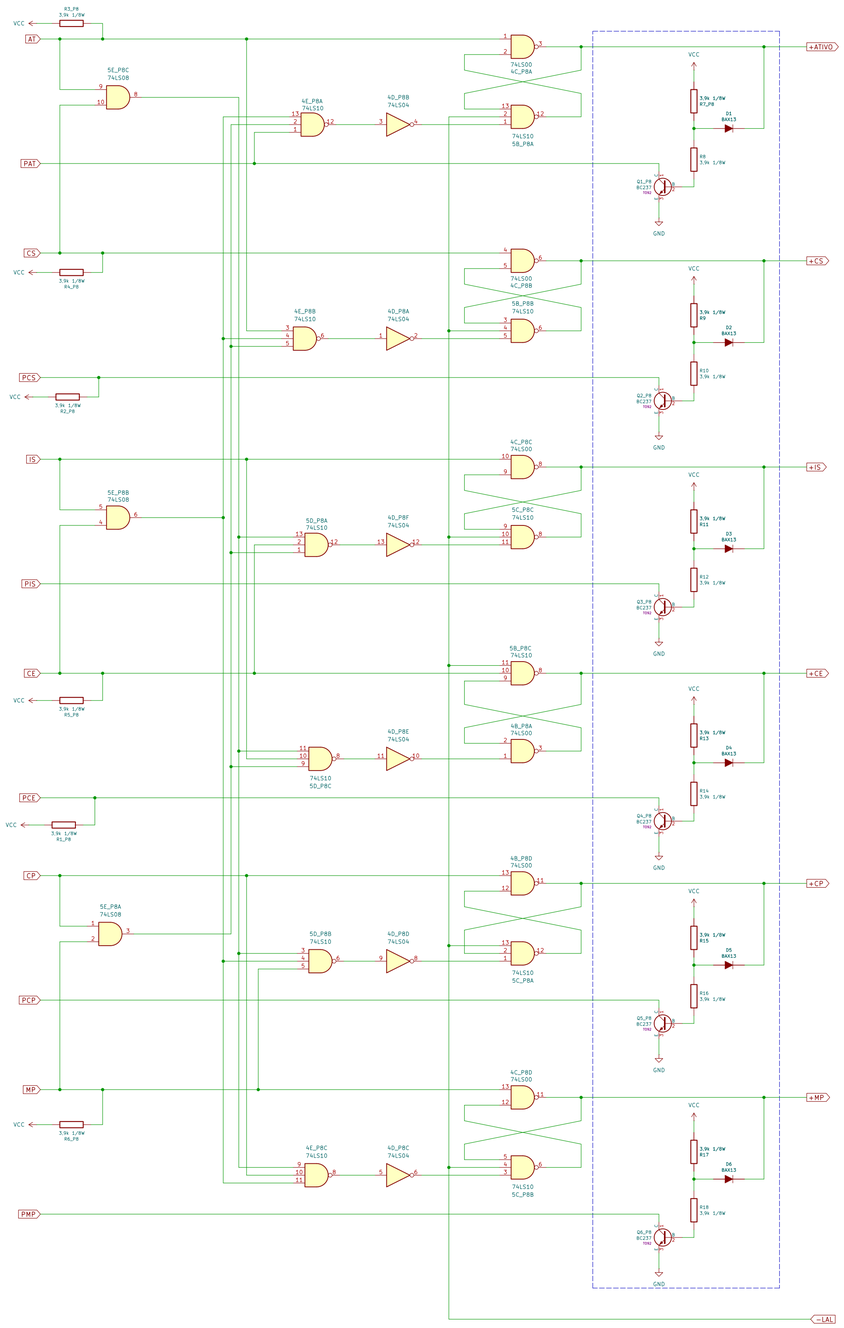


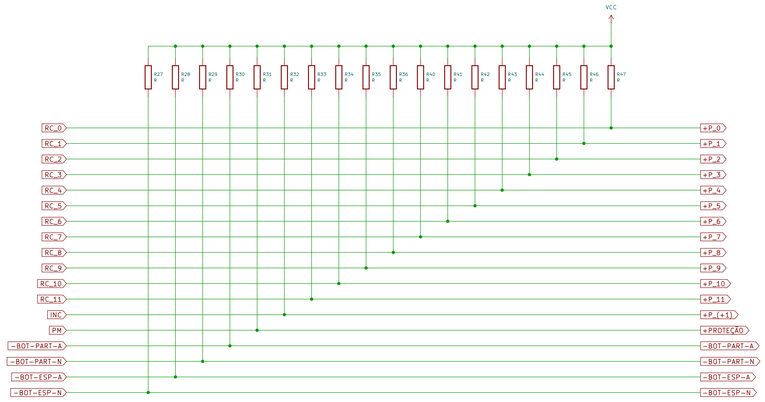


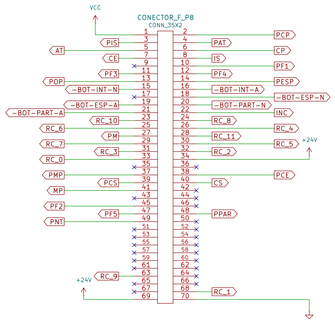
Sheet: /PRANCHA VII: PLACA MOSTRA REGISTRADORES/
File: PMR-1.kicad_sch

Title:

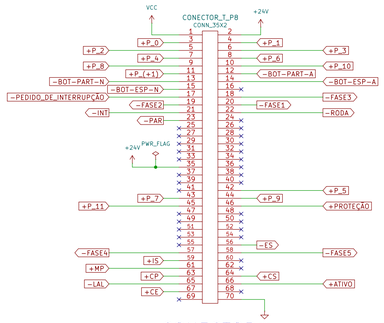
Size: A2	Date: 2 aug 2016	Rev:
KiCad E.D.A. 10.0.3		Id: 9/18



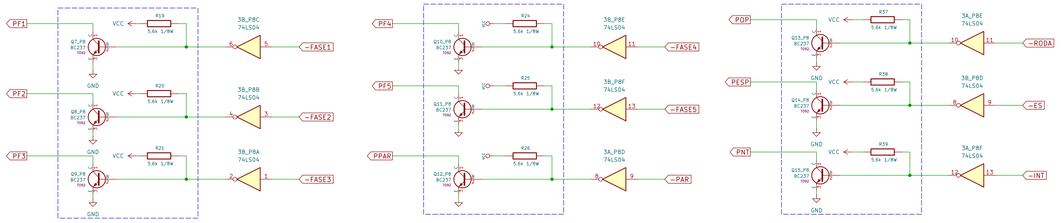


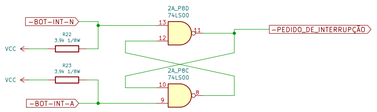


CONECTOR FRONTAL

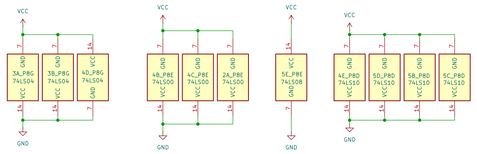


CONECTOR TRASEIRO





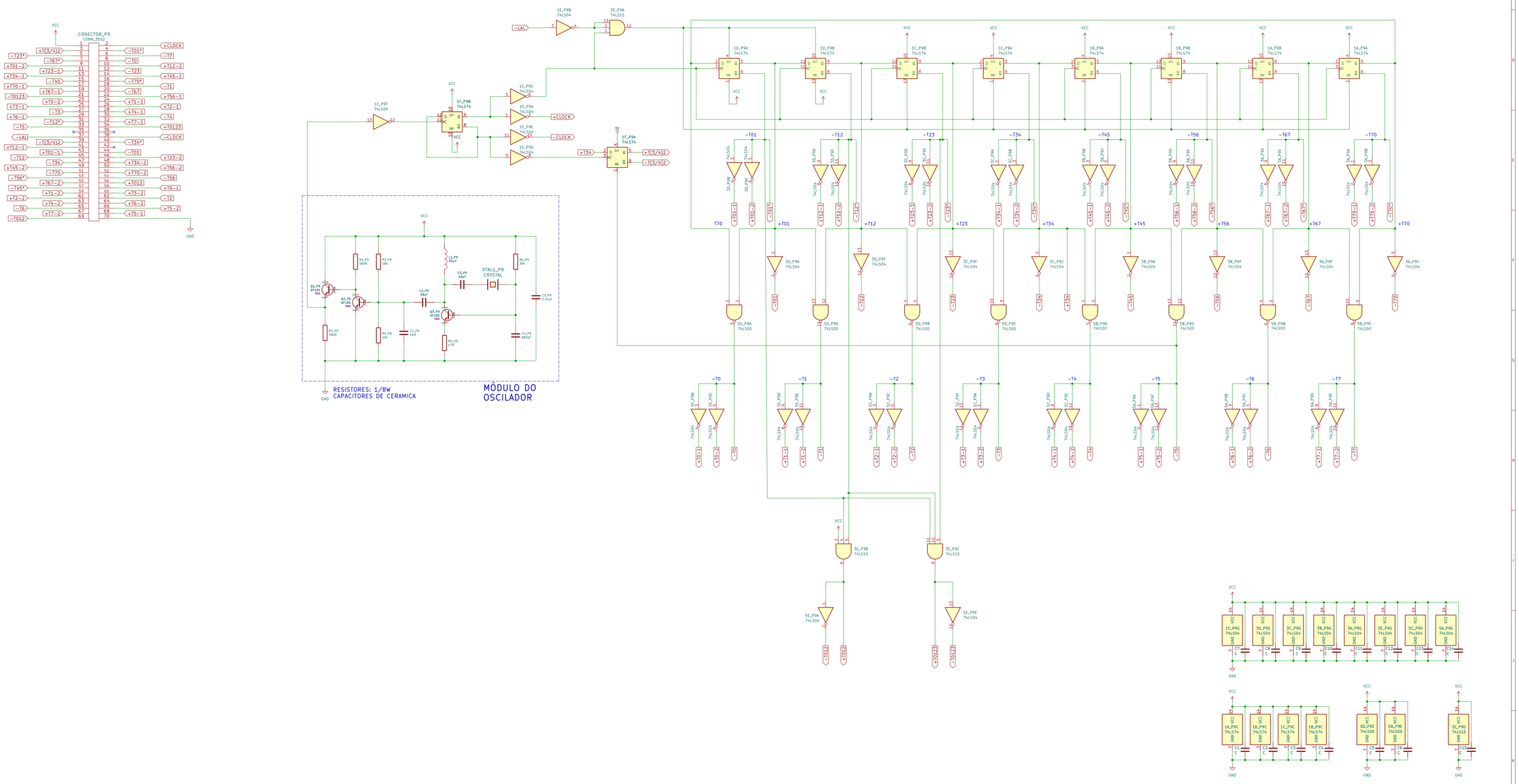
(!) Isso aparenta estar errado.
Há ainda 2 unidades disponíveis na pastilha 4B.
Por que não usá-las aqui?



Sheet: /PRANCHA VIII: PLACA CHAVES E PILOTO/
File: PCP-2.kicad_sch

Title:

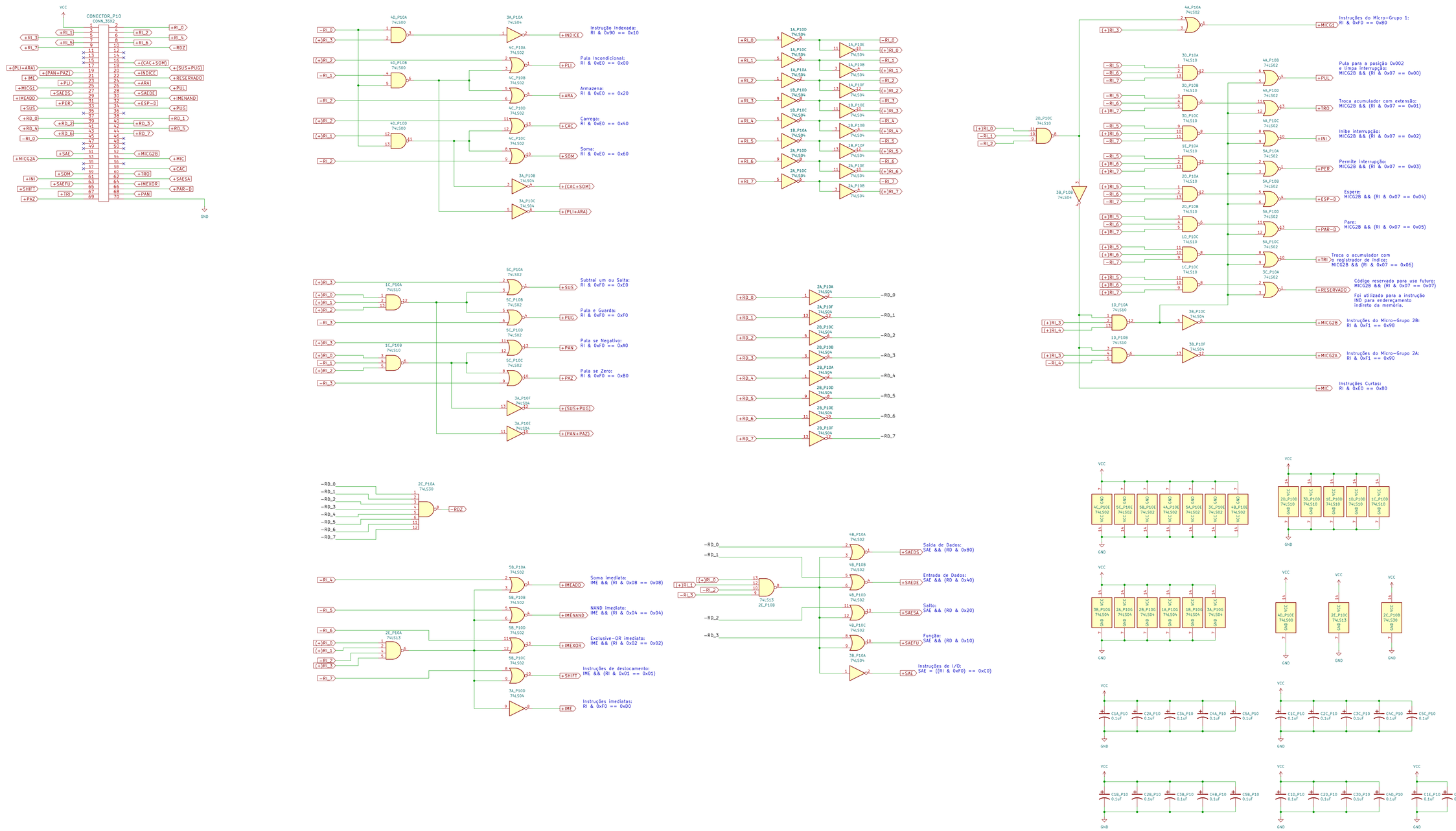
Size: A1	Date: 2 aug 2016	Rev:
KiCad E.D.A. 10.0.3		Id: 10/18



Sheet: /PRANCHA IX: RELÓGIO CENTRAL/
File: CRC-1.kicad_sch

Title:

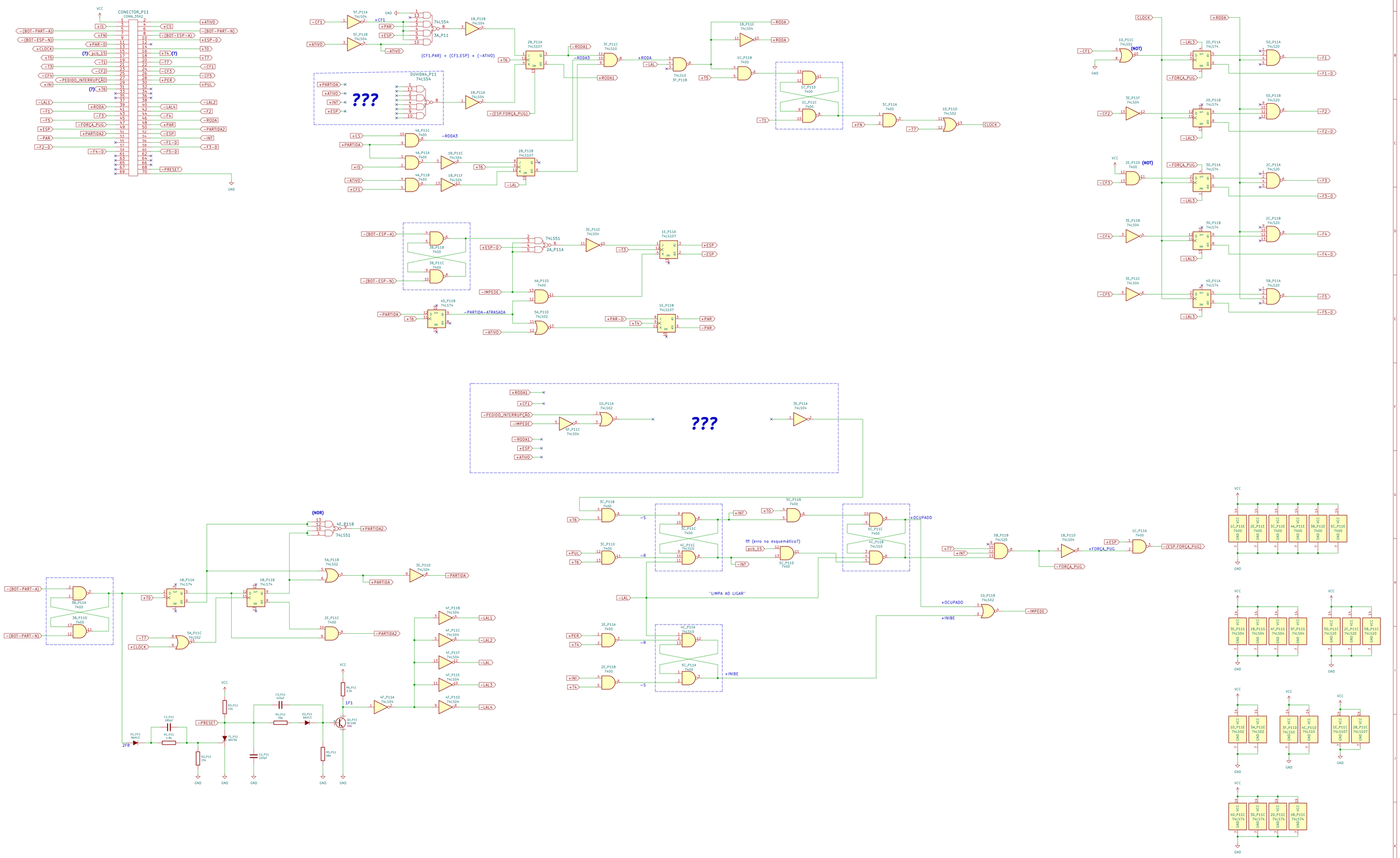
Size: A1	Date: 2 aug 2016	Rev:
KiCad E.D.A. 10.0.3		Id: 11/18

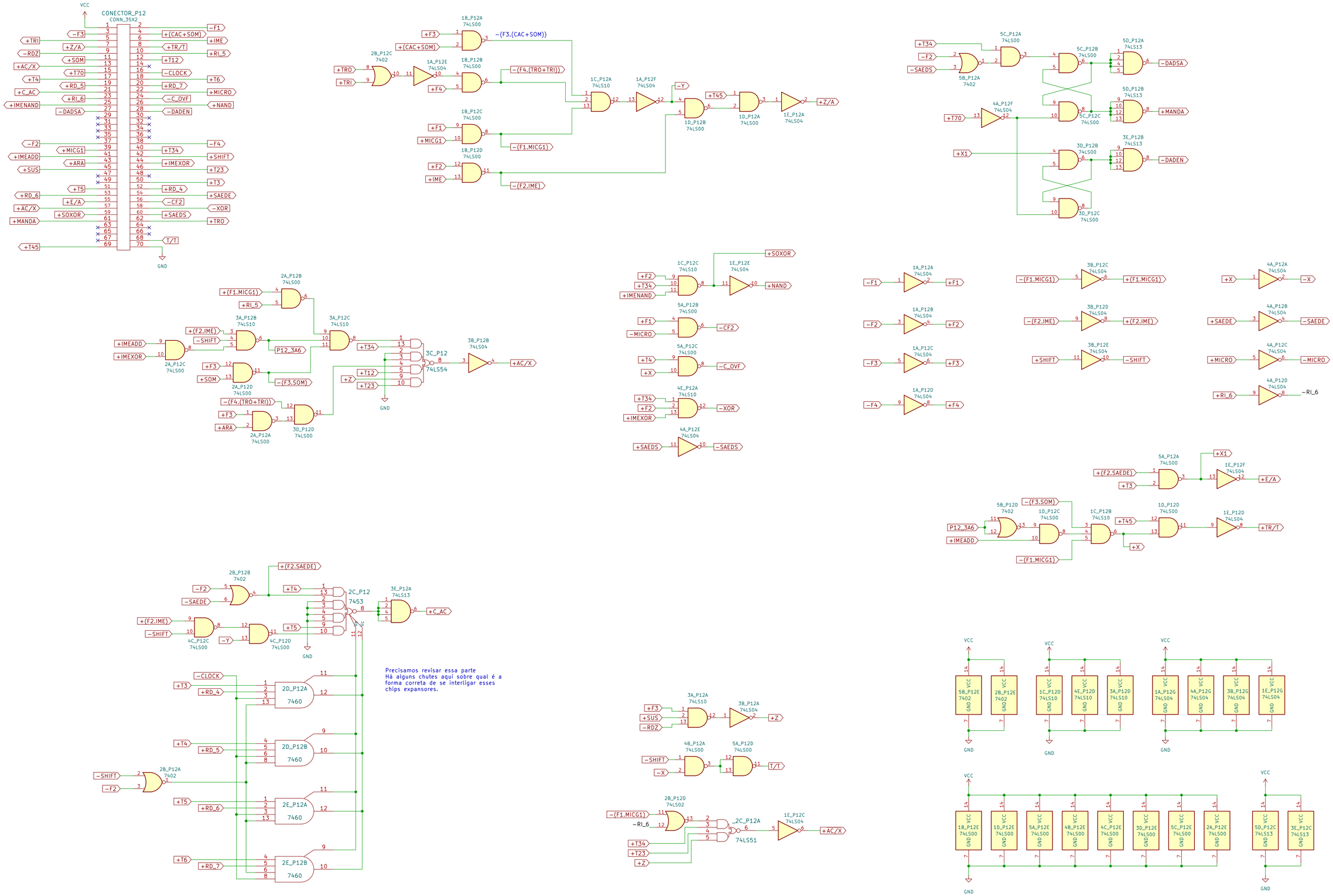


Sheet: /PRANCHA X: DECODIFICADOR/
File: CDE-3.kicad_sch

Title:

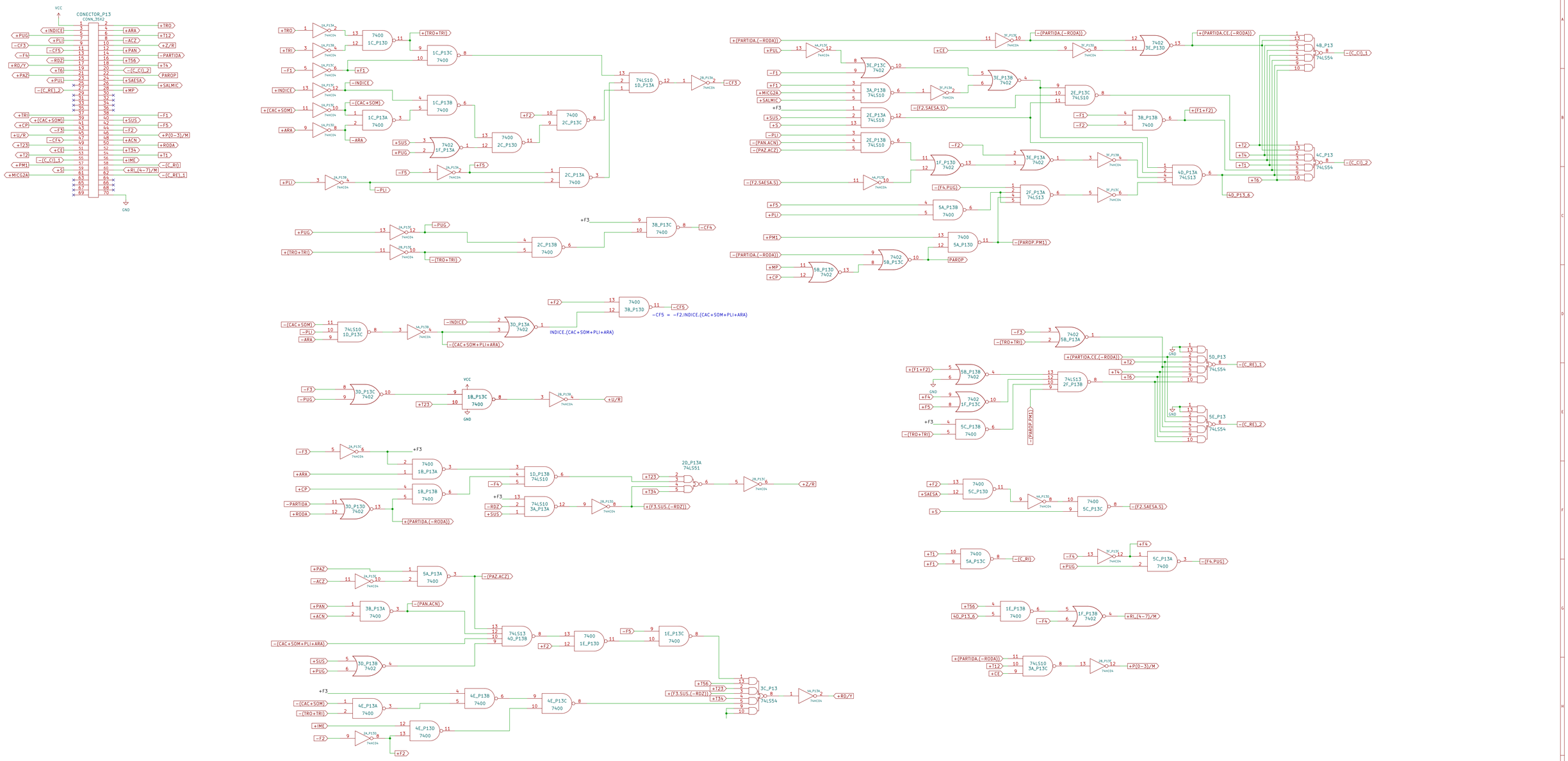
Size: A1	Date: 2 aug 2016	Rev:
KiCad E.D.A. 10.0.3		Id: 12/18


CONECTOR_P11
CONN_35X2
(NOR)
???
(CF1.PAR) + (CF1.ESP) + (−ATIVO)
"LIMPA AO LIGAR"
!!! (erro no esquemático?)
+OCUPADO
+INIBE
+FORÇA_PUG
−PRESET
−PARTIDA-ATRASADA
Sheet: /PRANCHA XI: CONTROLE DE ESTADO/
File: CCE-2.kicad_sch
Title:
Size: A1
Date: 2 aug 2016
KiCad E.D.A. 10.0.3
Rev:
Id: 13/18

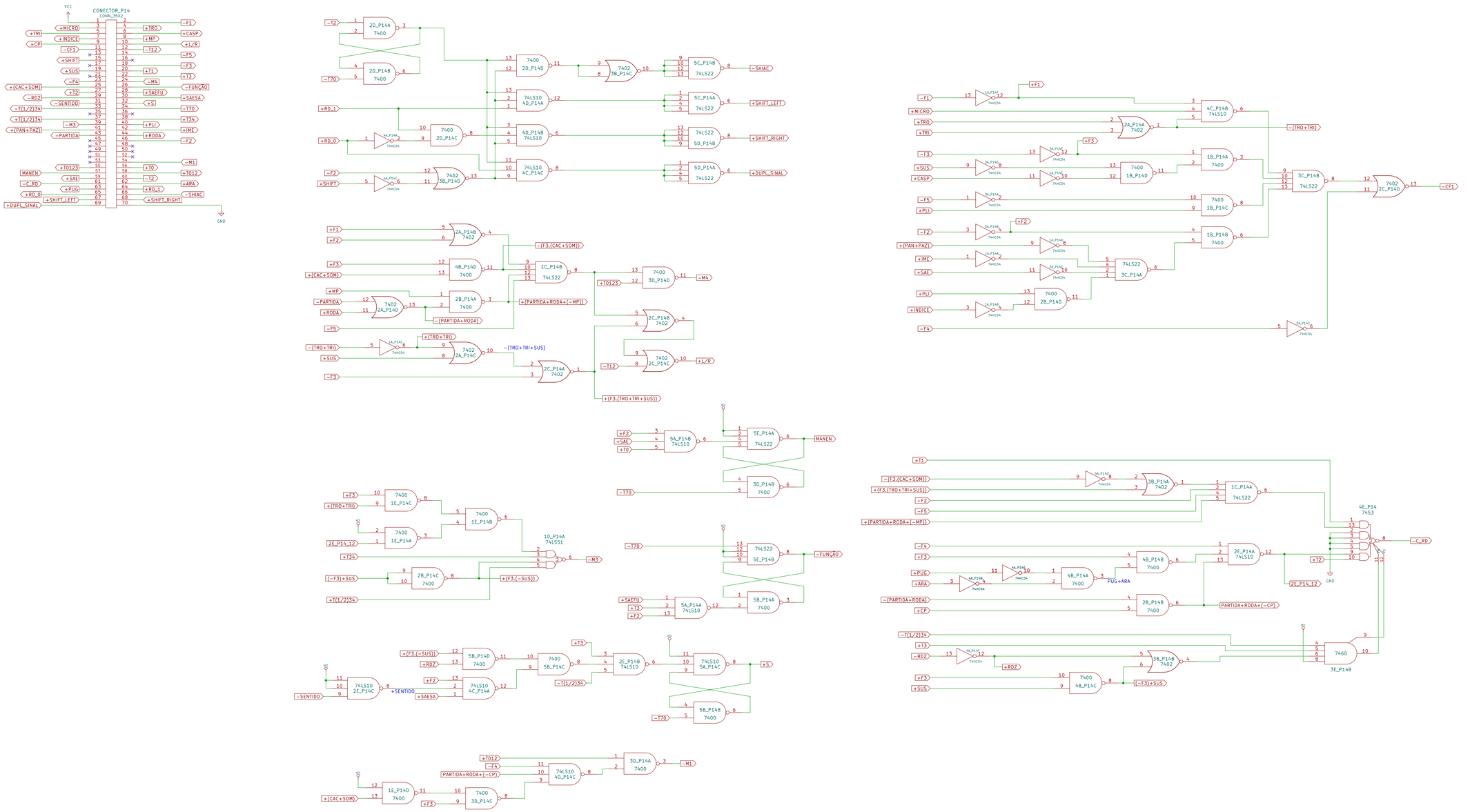


Precisamos revisar essa parte
Há alguns chutes aqui sobre qual é a
forma correta de se interligar esses
chips expansores.

Sheet: /PRANCHA XII: PLACA CONTROLE 1/		
File: CT1-4.kicad_sch		
Title:		
Size: A2	Date: 2 aug 2016	**Rev:**
KiCad E.D.A. 10.0.3		Id: 14/18



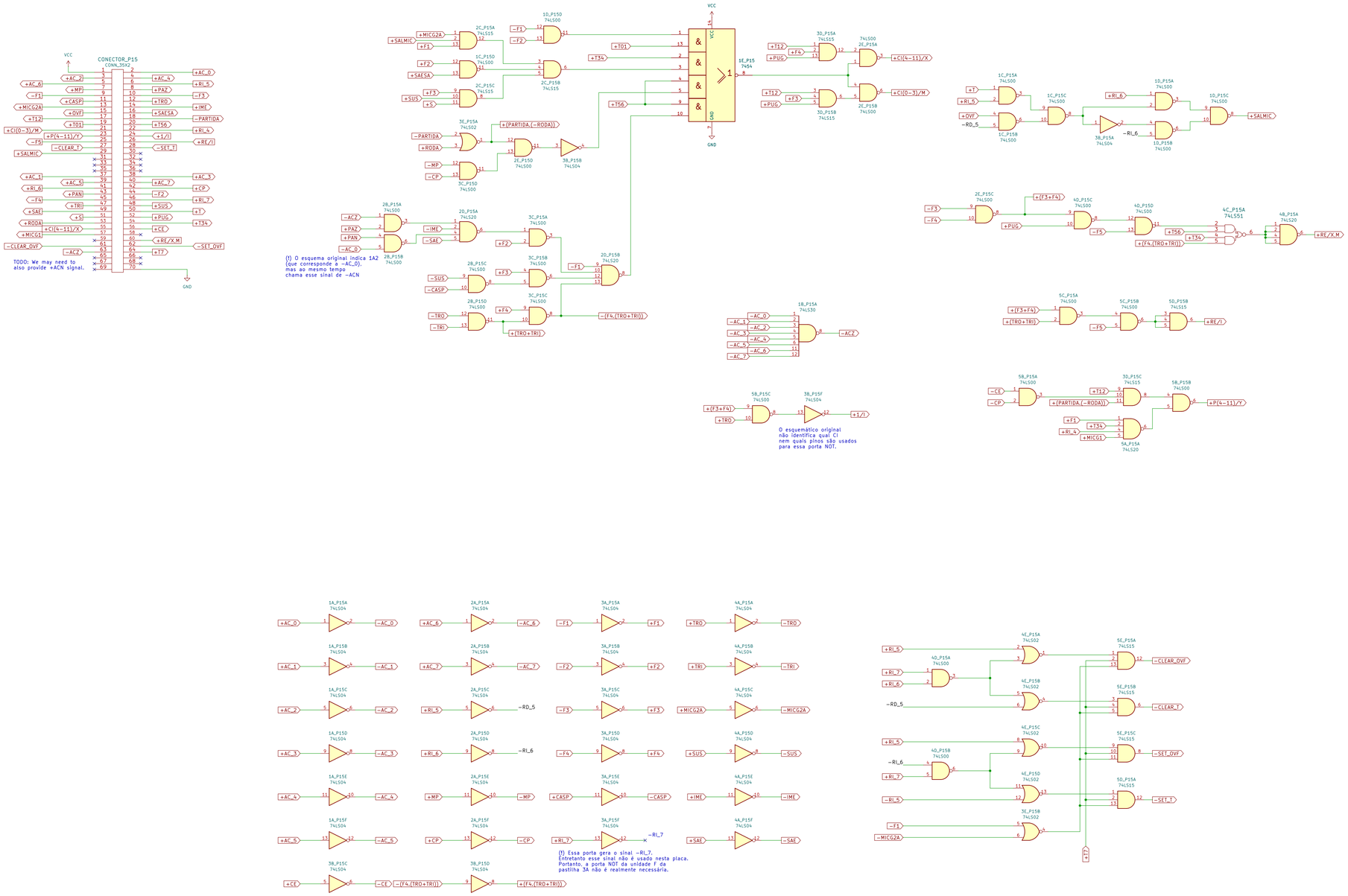
Sheet: /PRANCHA XIII: PLACA SINAIS CONTROLE/	
File: CT2-5.kicad_sch	
Title:	
Size: A1 Date: 2 aug 2016	**Rev:**
KiCad E.D.A. 10.0.3	Id: 15/18



Sheet: /PRANCHA XIV: PLACA DE CONTROLE/
File: CT3-6.kicad_sch

Title:

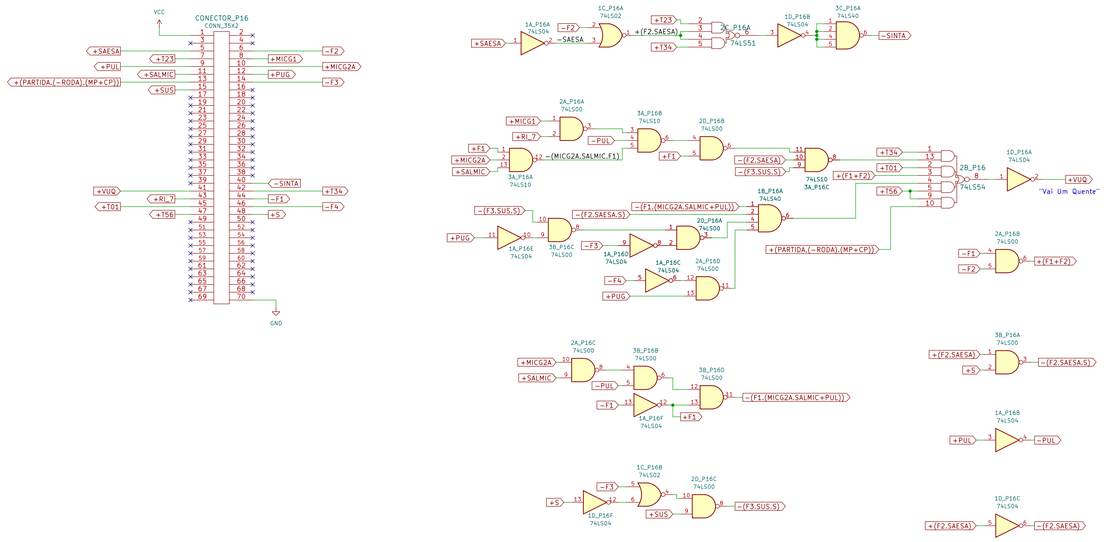
Size: A1	Date: 2 aug 2016	Rev:
KiCad E.D.A. 10.0.3		Id: 16/18

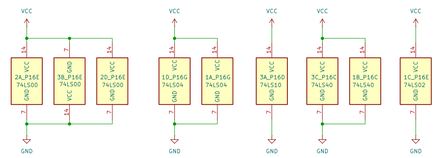


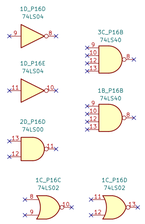
Sheet: /PRANCHA XV: SINAIS DE CONTROLE/
File: C14-7.kicad_sch

Title:

Size: A1	Date: 2 aug 2016	Rev:
KiCad E.D.A. 10.0.3		Id: 17/18







Re-desenho: Felipe Sanches
Projeto: Edson Freginl
Sinais de Controle
Computador de Controle - Patinho Feio
EPUSP LSD

Sheet: /PRANCHA XVI: SINAIS DE CONTROLE/
File: CTS-B.kicad_sch

Title: PRANCHA XVI

Size: A1	Date: 2 aug 2016	Rev:
KiCad E.D.A. 10.0.3		Id: 18/18